

Guarded periodic blocks and exact maximal order types of words below ω^2

A proposed solution to Altman's finite-alphabet sharpness question

Research report prepared with ChatGPT
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Abstract

For a finite poset P , consider all words of ordinal length less than ω^2 , ordered by strictly increasing subsequence embeddings that may increase labels in P . Write $n = |P|$ and let $j(P)$ count all downsets of P , including the empty one. This article gives a complete proposed proof that

$$o(s_{\omega^2}^F(P)) = \begin{cases} 1, & n = 0, \\ \omega^2, & n = 1, \\ \omega^{\omega^{n+j(P)-2}}, & n \geq 2. \end{cases}$$

In particular, the two-letter antichain has maximal order type ω^{ω^4} , answering the explicit sharpness question in Example 3.15 of Altman's March 2026 revision. The two-letter chain has type ω^{ω^3} . The main construction embeds every finite Cartesian power of an earlier stage into the next stage by synchronizing periodic ω -blocks. Proper downsets admit an excluded-letter guard; the universal block instead requires a limit-length padding argument. A more general classification for selected families of periodic tails isolates the exact exception in which naive block counting fails. Two routes from the product embeddings to the ordinal lower bound are given: the classical natural-product theorem, and an elementary hand-built linear extension of a Cartesian power that makes the main theorem independent of that import. Supplementary exact-length strata are computed, and they show explicitly why the answer cannot be assembled length by length. An exact symbolic embedding algorithm with canonical normal forms, negative controls for three invalid shortcuts, and a reproducible test suite accompany the proofs.

Status and scope. This is a new, unrefereed research argument, not a report of an independently certified result, and it has not been checked in a proof assistant. All proof steps beyond the explicitly stated classical maximal-order-type theorems are supplied below. The specific unresolved question was checked in the cited source; the claim of a solution rests on the argument given here and not on a claim of exhaustive literature coverage. The computer checks test symbolic embeddings, not infinite ordinal ranks. The theorem concerns *strictly increasing* embeddings, finite alphabet posets, and the strict length cutoff ω^2 ; it does not settle arbitrary alphabets or higher cutoffs.

Provenance. This report merges two independently produced research reports on the same question, order-types-below-omega-squared and guarded-periodic-blocks. Both reached the same formula by the same core mechanism: a finite generating alphabet for the upper bound, cofinal synchronization of periodic separators, an excluded-letter guard for proper downsets, and a limit-length guard for the universal one. That shared core is proved exactly once below. The two reports diverged at one step, the passage from the Cartesian-power embeddings to the ordinal lower bound, and both arguments are

kept: the classical natural-product route in Corollaries 5.3 and 6.4, and the elementary route of Section 6.4. Two different justifications of the monotone-surjection principle are likewise both recorded in Section 2.2. Material contributed by the second report, and absent from the first, includes the exact-length strata of Section 10, the canonical normal forms of Section 11.3, and the worked negative controls of Section 8.4.

Keywords. Well-quasi-order; maximal order type; transfinite word; recurrent downset; guarded periodic block; Higman ordering; finite downset lattice; ordinal arithmetic; natural product.

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1 The selected question and the result

Altman studies finite-image words of bounded transfinite length in [1]. Example 3.15, on printed page 10 of version 2, gives the upper bound ω^{ω^4} for a two-element antichain and explicitly leaves its sharpness unverified. The same example improves the chain's upper bound to ω^{ω^3} . The arXiv record checked for this note lists version 2, dated March 10, 2026, as the latest revision. Targeted searches did not locate a subsequent resolution; this is a limited literature check, not a guarantee of priority. No claim is made that the present literature check excludes every earlier formulation of the general finite-poset formula.

Related work. The classical maximal-linear-extension theory is due to de Jongh and Parikh [2], and the finite-word calculation is part of the de Jongh–Parikh–Schmidt theory [2, 3]; the underlying well-quasi-order assertion is Higman's lemma [4]. Recent work of Chopra and Pakhomov [5] studies the limit cutoff ω^ω through finite trees. The present article addresses the finite cutoff ω^2 and the explicitly identified two-letter example, so the two lines of work are complementary rather than overlapping.

This is an attractive problem to attack because the alphabet is tiny, all ω -tails have finite descriptions up to embedding, and the candidate ordinal is explicit. The difficult part is not finding more tail types. It is proving that the types already available contribute their full ordinal complexity after concatenation.

Let P be a finite poset. A *downset* is a subset $D \subseteq P$ such that $x \leq_P y \in D$ implies $x \in D$. Put

$$\mathcal{J}(P) = \{D \subseteq P : D \text{ is a downset}\}, \quad j(P) = |\mathcal{J}(P)|.$$

The empty set is counted in $j(P)$ throughout.

Theorem 1.1 (Finite-alphabet formula). *Let P be a finite poset with $n = |P|$ elements. With strictly increasing, label-respecting subsequence embedding and mutual embeddability quotiented out,*

$$o(s_{\omega^2}^F(P)) = \begin{cases} 1, & n = 0, \\ \omega^2, & n = 1, \\ \omega^{\omega^{n+j(P)-2}}, & n \geq 2. \end{cases} \quad (1.1)$$

The two small alphabet sizes are part of the theorem, not removable technicalities. In particular, substituting $n = 1$ into the third line would give an incorrect answer; Section 7.2 explains why the defect is structural.

Here $s_{\omega^2}^F(P)$ denotes words of length *strictly less than* ω^2 . Since P is finite, the finite-image superscript is automatic. The theorem is proved in Section 7 after two product embedding lemmas.

Corollary 1.2 (The selected binary cases). *For a two-element antichain A_2 and a two-element chain C_2 ,*

$$o(s_{\omega^2}^F(A_2)) = \omega^{\omega^4}, \quad o(s_{\omega^2}^F(C_2)) = \omega^{\omega^3}.$$

Proof. There are four downsets in A_2 and three in C_2 . Substitute $n = 2$ in (1.1). □

The proof establishes more than these two values. We may choose an arbitrary family of nonempty downsets and allow only the corresponding periodic ω -blocks. Unless the universal block is the sole allowed infinite block, the number of allowed block types determines the maximal order type exactly. The exceptional case is essential, not a technical inconvenience.

1.1 What the proof adds

The upper bound follows from a finite generating alphabet. The lower bound requires an additional construction. A new periodic block serves as a separator only when an embedding cannot hide it inside earlier blocks. Introducing downsets in inclusion order solves that problem. A finite guard then prevents a component from leaking into the next separator. At the universal separator no excluded letter exists, so a different device is needed: attach a cancellable tail of limit length. An infinite tail cannot fit into a bounded, hence finite, initial segment of an ω -block.

Together these observations give actual order embeddings

$$X^{r+1} \hookrightarrow Y \quad (r < \omega)$$

from each stage into the next. Classical natural-product arithmetic turns these embeddings into the sharp lower bound. Section 6.4 gives a second, elementary passage from the same embeddings to the same bound, using only an explicitly constructed linear extension of a Cartesian power; the main theorem therefore does not depend on importing the de Jongh–Parikh product formula.

What is claimed as the contribution. The normal-form encoding and the finite Higman maximal-order-type formula are established ingredients. The central contribution is the guarded product construction of Sections 5 and 6, which makes the generator-count upper bound sharp. The support-family classification and the finite-poset formula follow from that construction. The exact-length calculations of Section 10 are supplementary consequences, not the foundation of the main lower bound.

2 Definitions and classical inputs

2.1 Ordinal words and embeddability

A *word over P* is a function $u : \alpha \rightarrow P$, where α is an ordinal; its length is $\text{len}(u) = \alpha$. If $u : \alpha \rightarrow P$ and $v : \beta \rightarrow P$, write $u \preccurlyeq v$ when there is a strictly increasing map $f : \alpha \rightarrow \beta$ satisfying

$$u(\xi) \leq_P v(f(\xi)) \quad (\xi < \alpha).$$

No continuity or cofinality condition is imposed on f . Write $u \simeq v$ when $u \preccurlyeq v$ and $v \preccurlyeq u$. Concatenation is ordinal concatenation. The empty word ε has length zero. For lengths below ω^2 there are unique $k, m < \omega$ such that

$$\text{len}(u) = \omega k + m.$$

Embeddability is in general only a quasi-order. Distinct words, such as a^ω and aa^ω , may be equivalent. All maximal order types in this article refer to the quotient by $\simeq$.

Strictness of the index map is indispensable. Under a merely nondecreasing index map an infinite constant source could be collapsed onto a single target position, and every statement below would describe a different problem. The same remark explains why finite material preceding an infinite block can disappear from the *length* expression, by $k + \omega = \omega$, while its labels may still matter for embeddability.

Lemma 2.1 (Concatenation is monotone). *If $u_i \preccurlyeq v_i$ for $0 \leq i < r$, where r is finite, then $u_0 \cdots u_{r-1} \preccurlyeq v_0 \cdots v_{r-1}$.*

Proof. Translate the embedding on each component into the corresponding interval of the target concatenation. The intervals occur in the same order, so the union of these translated maps is strictly increasing and respects labels. $\square$

A finite concatenation of words of length below ω^2 still has length below ω^2 . In particular, all maps used later stay within the stated cutoff, even though the number of ω -blocks in their arguments is unbounded across different arguments.

2.2 Maximal order types

A quasi-order is a *well-quasi-order* if every infinite sequence $x_0, x_1, \dots$ contains $i < j$ with $x_i \leq x_j$. The maximal order type $o(X)$ is the largest ordinal type of a linear extension of the quotient poset. The existence of this maximum, its characterization by the tree of bad finite sequences, and the product theorem below are classical [2]. We use the following precise inputs.

Theorem 2.2 (Classical maximal-order-type calculus). *For well-quasi-orders X, Y :*

- (i) *An order embedding $X \hookrightarrow Y$ implies $o(X) \leq o(Y)$.*
- (ii) *An order-preserving surjection from Y onto X , including surjectivity up to quasi-order equivalence, implies $o(X) \leq o(Y)$.*

- (iii) For the componentwise Cartesian product, $o(X \times Y) = o(X) \otimes o(Y)$, where $\otimes$ is natural product.
- (iv) For a disjoint union, with no comparisons across the two parts, $o(X \amalg Y) = o(X) \oplus o(Y)$, where $\oplus$ is natural sum.
- (v) If P is a nonempty finite poset of cardinality n , its Higman word ordering satisfies

$$o(P^*) = H_n, \quad H_n := \omega^{\omega^{n-1}}. \quad (2.1)$$

The finite-word formula is part of the de Jongh–Parikh–Schmidt theory [2, 3]; its underlying well-quasi-order assertion is Higman’s lemma [4]. These are imported theorems; the present article does not claim new proofs of them. All later lower bounds are reduced to these inputs by explicit embeddings. Assertions (iii), (iv) and (v) are used as follows: (v) everywhere, (iii) in the main line of Corollaries 5.3 and 6.4 and, together with (iv), in the exact-length calculations of Section 10. The alternative route of Section 6.4 shows that (iii) can be dropped from the main theorem, though not from Section 10.

Assertion (ii) admits two elementary explanations, and both are recorded because they illuminate different sides of the statement.

First argument, on bad sequences. Choose a preimage for each element of the target quotient. If a target sequence is bad, its sequence of chosen preimages is bad, because any comparison in the source would project to a comparison in the target. This embeds the target bad-sequence tree into the source tree. The same argument shows the target is a well-quasi-order whenever the source is.

Second argument, on fibres. Choose a linear extension of the target X of type $o(X)$ and order the fibres of the surjection consecutively according to that extension, then linearize each fibre arbitrarily. Because the map is order preserving, no comparison of the source is violated by this arrangement: a source comparison projects to a target comparison, which is respected by the chosen ordering of fibres, and comparisons inside a single fibre are respected by its own linearization. The result is a linear extension of Y of type at least $o(X)$, so $o(X) \leq o(Y)$. The first argument also yields the well-quasi-order assertion; the second exhibits a concrete linear extension and is what the exact-length upper bounds of Section 10 reuse.

2.3 The ordinal arithmetic used here

For ordinals in Cantor normal form, natural sum $\oplus$ adds coefficients at equal exponents, and natural product $\otimes$ multiplies the resulting formal polynomials, using natural sum on exponents. In particular,

$$(\omega^\alpha) \otimes (\omega^\beta) = \omega^{\alpha \oplus \beta}.$$

Consequently, for finite $N \geq 1$ and $q \geq 1$,

$$H_N^{\otimes q} = \omega^{\omega^{N-1}q}, \quad (2.2)$$

where the multiplication by q on the right is ordinary ordinal multiplication by a positive integer. Continuity of ordinal exponentiation gives

$$\sup_{q \geq 1} H_N^{\otimes q} = \omega^{\sup_{q \geq 1} \omega^{N-1}q} = \omega^{\omega^N} = H_{N+1}. \quad (2.3)$$

Equation (2.3) is the entire ordinal calculation behind the main induction.

A companion identity, using *ordinary* ordinal exponentiation only, will be needed for the alternative route of Section 6.4: for $q \geq 1$,

$$\sup_{r \geq 1} H_q^r = \sup_{r \geq 1} \omega^{\omega^{q-1} \cdot r} = \omega^{\omega^q} = H_{q+1}. \quad (2.4)$$

Ordinary powers and natural powers must not be interchanged in general, and (2.2)–(2.3) and (2.4) are emphatically not typographical variants of one another. In (2.2) the exponent $\otimes q$ is a natural power of H_N , computed by natural multiplication of Cantor normal forms; in (2.4) the exponent r is an ordinary ordinal power, computed by iterated ordinary multiplication. The two displays happen to have the same supremum, which is exactly why the two routes to the lower bound reach the same ordinal, and each display states exactly the special case that its own route uses. Neither may be substituted for the other inside a proof.

3 Periodic tails and the finite generating alphabet

3.1 One periodic block for each nonempty downset

For a nonempty downset $D = \{d_0, \dots, d_{t-1}\}$, fix a periodic word

$$B_D = (d_0 \cdots d_{t-1})^\omega. \quad (3.1)$$

The chosen enumeration is immaterial up to $\approx$: every element of D appears infinitely often in every such representative.

Lemma 3.1 (Comparison of periodic blocks). *For nonempty downsets D, E and a letter $a \in P$,*

$$a \preccurlyeq B_D \iff a \in D, \quad B_D \preccurlyeq B_E \iff D \subseteq E.$$

Every nonempty tail of B_D is equivalent to B_D , and a letter a embeds into every tail of B_D exactly when $a \in D$. Consequently, if every letter of a finite word p lies in D , then

$$pB_D \approx B_D. \quad (3.2)$$

Proof. If $a \leq_P d$ for a letter $d \in D$, downward closure gives $a \in D$. Conversely, $a \in D$ supplies a matching occurrence in B_D . If $B_D \preccurlyeq B_E$, applying the same observation to an occurrence of each $d \in D$ gives $D \subseteq E$. If $D \subseteq E$, recursively match the letters of B_D to successive identical occurrences in B_E . There is always a later occurrence, so the recursion gives an embedding. The identical recursive construction works in any nonempty tail of a periodic block; the reverse embedding into the full block is inclusion. For (3.2), place the finitely many letters of p in an initial portion of B_D , which is possible because each of them lies in D and every element of D recurs, and embed the remaining copy of B_D into the tail that is left. The reverse embedding simply omits p . $\square$

Note that (3.2) holds for an arbitrary nonempty downset, not only for $D = P$. The special case $D = P$, where p may be any finite word at all, is isolated later as (6.1).

For a nonempty subset $R \subseteq P$, repeating R or repeating its downset $\downarrow R$ gives equivalent ω -words. In the less immediate direction, each $x \in \downarrow R$ can be matched to an element $r \in R$ above it, and each such r recurs infinitely often.

Remark 3.2 (Repeating only the maximal elements). Let $\text{Max } D$ denote the set of maximal elements of a nonempty downset D . Applying the previous paragraph with $R = \text{Max } D$, whose downset is D , gives

$$B_{\text{Max } D} \approx B_D.$$

Some presentations of this material define the periodic representative by repeating $\text{Max } D$ rather than all of D . The two conventions are interchangeable everywhere below, and we keep the all-elements block B_D of (3.1) as the official definition. Where a proof below inspects the letters of a block, it only ever uses that the letters lie in D and that every element of D recurs, which both representatives satisfy.

3.2 Reduction of arbitrary words

For a word $u : \omega \rightarrow P$ put

$$\text{Rec}(u) = \{a \in P : a \text{ occurs infinitely often in } u\}, \quad I(u) = \downarrow \text{Rec}(u). \quad (3.3)$$

The set $\text{Rec}(u)$ is nonempty because P is finite and u is infinite. Its downward closure $I(u)$, and not the literal recurrent set, is the correct invariant here: an embedding may increase labels, so what a block can absorb is determined by the downset generated by its recurrent letters. We call $I(u)$ the *recurrent downset* of u .

The following periodic-tail reduction is standard; compare [1, Propositions 3.3–3.4]. A proof is included to specify exactly which equivalence is needed.

Lemma 3.3 (Finite prefix and recurrent tail). *Every word $u : \omega \rightarrow P$ is equivalent to vB_D for some finite word $v \in P^*$ and some nonempty downset D ; one may take $D = I(u)$.*

Proof. Let $R = \text{Rec}(u)$ be the set of labels occurring infinitely often in u . Since P is finite and u is infinite, R is nonempty. Every label outside R occurs only finitely often. Choose an index after all such occurrences, and write $u = vt$, where v is finite and t uses only labels from R . Every element of R occurs infinitely often in t .

Put $D = \downarrow R$. To embed t into B_D , successively choose later identical occurrences of its labels. To embed B_D into t , for each requested label $d \in D$ choose an $r \in R$ with $d \leq_P r$, and then choose an occurrence of r later than all previous choices. Infinitely many occurrences of r make this possible. Thus $t \simeq B_D$, and monotonicity of concatenation gives $u \simeq vB_D$. $\square$

The prefix can be normalized further, and the strengthened form is what the canonical representatives of Section 11.3 and the exact-length computations of Section 10 require.

Lemma 3.4 (One-block normal form). *Every word $u : \omega \rightarrow P$ is equivalent to pB_D with $D = I(u)$, where the finite word p may in addition be required to be empty or to end in a letter outside D .*

Proof. Each letter outside D occurs only finitely often in u , and there are finitely many letters, so either no letter outside D occurs, in which case put $p = \varepsilon$, or there is a last such occurrence, in which case let p be the prefix ending at that occurrence. In both cases write $u = pt$ with every label of t inside D .

Every element of D still recurs in t : given $d \in D$ there is, by $D = \downarrow \text{Rec}(u)$, a recurrent $b \geq_P d$, and only finitely many positions were removed with p , so b still recurs in t . Greedy matching of the labels of t to dominating occurrences gives $t \preceq B_D$, and matching each requested $d \in D$ to a later occurrence of such a b gives $B_D \preceq t$. Hence $t \simeq B_D$, and concatenation gives $u \simeq pB_D$. $\square$

Proposition 3.5 (Finite representation). *Every word of length less than ω^2 is equivalent to a finite concatenation of one-letter words and blocks B_D , with $\emptyset \neq D \in \mathcal{J}(P)$. More precisely, a word of length $\omega r + k$ is equivalent to an expression*

$$p_0 B_{D_0} p_1 B_{D_1} \cdots p_{r-1} B_{D_{r-1}} p_r, \quad (3.4)$$

in which every p_i is a finite word, every p_i with $i < r$ is empty or ends in a letter outside D_i , and p_r has length k .

Proof. Split a word of length $\omega r + k$ into its r consecutive canonical intervals of length ω and its final finite interval of length k . Apply Lemma 3.3 to each infinite interval and concatenate the resulting representatives; applying Lemma 3.4 instead yields the normalized prefixes of (3.4). $\square$

By itself this is a representation theorem, not a uniqueness theorem: without the extra clause in (3.4), different finite expressions represent equivalent words, for example $aB_{\downarrow a} \simeq B_{\downarrow a}$. Such absorption is precisely why a finite generating alphabet gives only an upper bound without further work. Uniqueness is nevertheless recoverable once the prefixes are normalized as in (3.4); this is carried out, with an explicit stack-based procedure, in Section 11.3.

The construction is existential for an arbitrary infinite input word, since it uses knowledge of which letters recur. The algorithm of Section 11 takes finite expressions as input; it is not an algorithm for discovering recurrence in an arbitrary stream.

Definition 3.6 (Selected families). For $\mathcal{F} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$, let $\mathcal{W}(P, \mathcal{F})$ be the quasi-order of all finite concatenations of letters in P and periodic blocks B_D with $D \in \mathcal{F}$, using actual ordinal-word embeddability. Finite expressions are identified up to mutual embeddability when maximal order type is taken.

In particular, $\mathcal{W}(P, \emptyset) = P^*$, and $\mathcal{W}(P, \mathcal{J}(P) \setminus \{\emptyset\})$ has the same quotient order as $s_{\omega^2}^F(P)$ by Proposition 3.5.

Remark 3.7 (An intrinsic description of $\mathcal{W}(P, \mathcal{F})$). Definition 3.6 is syntactic, but the resulting class of words has an intrinsic description: up to equivalence, $\mathcal{W}(P, \mathcal{F})$ consists exactly of those words below ω^2 each of whose canonical ω -blocks has its recurrent downset in $\mathcal{F}$. A finite prefix cannot change the recurrent downset of a block, which gives one inclusion, and Lemma 3.4 gives the converse by rewriting each block as pB_D with $D = I(u) \in \mathcal{F}$.

Proposition 3.8 (Atom-count upper bound). *For $n = |P| \geq 1$ and $m = |\mathcal{F}|$,*

$$o(\mathcal{W}(P, \mathcal{F})) \leq H_{n+m}. \quad (3.5)$$

Moreover, $\mathcal{W}(P, \mathcal{F})$ is a well-quasi-order.

Proof. Take a discrete alphabet of $n + m$ symbols: one for each finite letter and one for each selected periodic block. Evaluate a finite symbol string by ordinal concatenation. If one string is a subsequence of another, the corresponding evaluated word embeds into the other by Lemma 2.1. Evaluation is surjective onto the represented quasi-order. Apply the finite-word formula and monotone-surjection principle in Theorem 2.2. $\square$

Using a discrete alphabet deliberately ignores extra comparisons among the atoms. This does not weaken the desired upper bound: every poset with the same positive finite cardinality has the same finite-word maximal order type. No order-reflection claim is made for the evaluation map.

4 Cofinal localization and synchronization

The following lemma concerns *displayed target tokens*. A finite word may contain many one-letter tokens adjacent to each other, and a token expression need not be a canonical decomposition of its ordinal length. Nevertheless, each displayed B_E occupies a convex interval of order type ω , and there are only finitely many displayed tokens.

Lemma 4.1 (Cofinal localization). *Suppose a periodic block B_D embeds into a finite token expression v . Then a tail of its image lies in one displayed target block B_E , is cofinal in that block, and satisfies $D \subseteq E$. If a source position occurs after the whole source block B_D , its image occurs after the whole target block B_E .*

Proof. Number the target tokens in their order of occurrence. As the source positions $0, 1, 2, \dots$ increase, the indices of the tokens containing their images form a nondecreasing sequence with only finitely many possible values. It is eventually constant. The final token cannot be a one-letter token, because the embedding is injective. It is therefore some B_E . The image contains infinitely many positions of that token, so it is unbounded in its order type ω .

Every $d \in D$ still occurs in the source tail. Each such occurrence maps to a label in E above d . Downward closure gives $d \in E$, proving $D \subseteq E$. Finally, the image of any subsequent source position must lie above every image point of B_D . Since those points are cofinal in the target B_E , no position of that target block remains available. $\square$

Remark 4.2 (A block may straddle earlier blocks). The lemma does not say that the entire source block lands in a single target block. An initial finite part can be placed earlier. Only the tail is localized. All subsequent proofs use this weaker, correct statement. Under the stronger hypothesis that source and target have the *same* number of canonical ω -blocks, whole-block containment does hold; that is Lemma 10.2 below, which is used only in Sections 10 and 11.3. The two statements are compatible because their hypotheses differ, and nothing in Lemma 10.2 retracts the present remark.

Remark 4.3 (The role of explicit segments). A displayed token B_E inside an expression need not begin at a canonical ω -block boundary of the word it denotes: if a finite word precedes it, its pure periodic segment starts at a finite offset inside the corresponding canonical block. That segment still has length ω , is cofinal in the block, and ends at the next canonical boundary; this is the ordinal identity $k + \omega = \omega$ again. Throughout the arguments below, “inside a displayed separator” therefore always means inside that *explicit pure periodic segment*, never inside the finite prefix that happens to precede it. This is a statement about the *target* side, complementary to Remark 4.2, which is about a *source* block straddling earlier target blocks.

Lemma 4.4 (Equal-count synchronization). *Let D be a nonempty downset, and suppose an old family $\mathcal{G}$ satisfies*

$$D \not\subseteq E \quad \text{for every } E \in \mathcal{G}. \quad (4.1)$$

Consider two token expressions with exactly r displayed copies of B_D , all other infinite tokens belonging to $\mathcal{G}$. Under an embedding of the first expression into the second, the tail of the i th displayed source B_D is cofinal in the i th displayed target B_D , for $1 \leq i \leq r$.

Proof. By Lemma 4.1, each displayed source B_D requires a target infinite block whose downset contains D . Condition (4.1) excludes all old target blocks. Its cofinal destination is therefore one of the r displayed new target blocks. Successive source new blocks have strictly increasing cofinal destination indices: after the first has used a target block cofinally, any later source position must be beyond it. A strictly increasing injection from $\{1, \dots, r\}$ into itself is the identity. This proves the claim. $\square$

Corollary 4.5 (Location between synchronized boundaries). *In the situation of Lemma 4.4, write the source expression as $b_0 B_D b_1 B_D \cdots B_D b_r$ and the target as $d_0 B_D d_1 B_D \cdots B_D d_r$, with all b_i, d_i in $\mathcal{W}(P, \mathcal{G})$. Then, for each i :*

- (a) *if $i > 0$, every image point of b_i occurs at or after the end of the i th displayed target separator;*
- (b) *if $i < r$, every image point of b_i occurs before the image of the first point of the $(i + 1)$ st source separator, hence before the end of the $(i + 1)$ st displayed target separator;*
- (c) *b_r embeds into d_r .*

Proof. Every point of b_i follows every point of the preceding source separator. That separator's image is cofinal in the corresponding target separator by Lemma 4.4, so no target position of that separator remains available, giving (a). Every point of b_i precedes the next source separator, whose first image point lies before the end of the corresponding target separator, again by cofinality; strict monotonicity gives (b). After the last target separator, the only remaining target material is d_r , which with (a) gives (c). $\square$

Part (b) still permits a finite tail of b_i to enter the next target separator. The guards of Sections 5 and 6 eliminate precisely that possibility.

To arrange (4.1), introduce a finite collection of downsets in an inclusion-respecting order. Ordering them by cardinality, with arbitrary tie breaking, suffices: a new set cannot be contained in an earlier, distinct set of no larger size. The full downset P , when present, is last.

5 Proper markers: finite guards give product embeddings

Lemma 5.1 (Right cancellation of a final letter). *For arbitrary ordinal words u, v and any fixed letter c ,*

$$uc \preceq vc \implies u \preceq v.$$

Proof. The image of the final source letter lies at or before the final target letter. Every position belonging to u maps strictly before that image, hence inside the target prefix v . Restrict the embedding. $\square$

Theorem 5.2 (Proper-marker product embedding). *Let D be a nonempty proper downset of P , and let $\mathcal{G}$ satisfy (4.1). Choose $c \in P \setminus D$ and put $X = \mathcal{W}(P, \mathcal{G})$, $Y = \mathcal{W}(P, \mathcal{G} \cup \{D\})$. For every $r \geq 1$, the map*

$$\Phi_r(u_0, \dots, u_r) = u_0 c B_D u_1 c B_D \cdots u_{r-1} c B_D u_r \tag{5.1}$$

is an order embedding $X^{r+1} \hookrightarrow Y$.

Proof. Order preservation follows by concatenating componentwise embeddings and identity maps on the separators. It remains to prove reflection.

Suppose $\Phi_r(u_0, \dots, u_r) \preceq \Phi_r(v_0, \dots, v_r)$. By Lemma 4.4, the i th source separator B_D has image tail cofinal in the i th target separator B_D . Thus, for $i > 0$, every image point of u_i and of its subsequent guard lies after the i th target separator. For $i = 0$ there is no preceding separator and no lower-bound restriction to impose.

Fix $i < r$. The source guard c immediately following u_i precedes the next source separator. Some point of that next separator maps into the corresponding next target separator. Therefore the source guard's image is before the end of that target separator. It cannot be in that separator: if it were matched to a label $d \in D$, then $c \leq_P d$ would imply $c \in D$, contrary to the choice of c . Its image consequently lies in the target interval $v_i c$ before the separator. Every point of u_i precedes the guard, so its image lies in v_i , not in the final guard of $v_i c$. We obtain $u_i \preceq v_i$.

After the last source separator, cofinality forces the entire source suffix u_r to map into the target suffix v_r . Hence $u_r \preceq v_r$ as well. All components compare in the product order. Reflection and preservation imply an order embedding after taking equivalence classes. $\square$

The guard is allowed to map into the interior of v_i ; it need not match the explicitly displayed target guard. The proof only needs the fact that it lies no later than that guard. This is important when old components already contain many occurrences of c or old infinite blocks.

Corollary 5.3 (One proper marker raises the level). *Under the hypotheses of Theorem 5.2, if $o(X) = H_N$ and $n + |\mathcal{G}| = N$, then $o(Y) = H_{N+1}$.*

Proof. For every $r \geq 1$, product embedding and the classical product theorem give $o(Y) \geq H_N^{\otimes(r+1)}$. Taking the supremum and using (2.3) yields $o(Y) \geq H_{N+1}$. Proposition 3.8 gives the reverse inequality. $\square$

6 The universal marker: limit padding replaces a guard

For $D = P$ there is no excluded letter. Any fixed finite word z is absorbed by the universal periodic block:

$$zB_P \simeq B_P. \quad (6.1)$$

Indeed, the countable sequence zB_P can be matched greedily into B_P ; every requested label reappears infinitely often. Thus attaching a finite guard before B_P cannot preserve arbitrary component information. The remedy is to end the component with a cancellable *infinite* tail.

6.1 A cancellable map into limit lengths

Lemma 6.1 (Limit padding). *Let E be a nonempty proper downset of P , and choose $c \notin E$. The map*

$$G(u) = ucB_E \quad (6.2)$$

is order preserving and order reflecting on arbitrary words of length below ω^2 . Every $G(u)$ has nonzero limit length, of the form ωk with $k \geq 1$. If B_E is allowed in a family, G maps that family's represented word order into itself.

Proof. Monotonicity is immediate. Suppose $ucB_E \preceq vcB_E$. The source occurrence of c cannot map into the final target B_E , by downward closure and $c \notin E$. Thus it maps into vc , and the restriction to u maps into v , by the same final-letter argument as Lemma 5.1. Hence $u \preceq v$.

If $\text{len}(u) = \omega k + m$, then

$$\text{len}(ucB_E) = (\omega k + m) + 1 + \omega = \omega(k + 1),$$

which is a nonzero limit ordinal. The family assertion follows from the explicit concatenation. $\square$

Lemma 6.2 (No infinite tail fits below a finite bound). *Let u have nonzero limit length below ω^2 . Suppose an embedding maps all of u to positions preceding a fixed position η inside a target interval of order type ω . Then no position of u can map inside that interval.*

Proof. If a source position ξ mapped into the interval before η , there would be only finitely many target positions between its image and η . But a nonzero limit ordinal has infinitely many positions after each of its positions: otherwise the ordinal would have a last position. The infinitely many source positions after ξ would have to inject into that finite target interval. This is impossible. $\square$

The position η need not be the first position of the target ω -block. Any point of it gives a finite bound. Nor is it necessary for the next source marker to begin inside that block: one image point of its cofinal tail suffices.

The lemma must not be misread as saying that a word of limit length never embeds into a universal periodic word. It certainly may; B_P alone embeds into B_P . The obstruction exploited below is different, and it is created by the surrounding configuration: a further synchronized source separator is required to begin at an actual target position before the end of that same target block, so the padded component and that separator would have to share one interval of order type ω with only finitely many positions between them.

6.2 The product embedding

Theorem 6.3 (Universal-marker product embedding). *Let $\mathcal{G}$ consist of nonempty proper downsets of P , with $\mathcal{G} \neq \emptyset$. Choose $E \in \mathcal{G}$ and the map G from Lemma 6.1. Put $X = \mathcal{W}(P, \mathcal{G})$ and $Y = \mathcal{W}(P, \mathcal{G} \cup \{P\})$. For each $r \geq 1$,*

$$\Psi_r(u_0, \dots, u_r) = G(u_0)B_P G(u_1)B_P \cdots G(u_{r-1})B_P u_r \quad (6.3)$$

is an order embedding $X^{r+1} \hookrightarrow Y$.

Proof. Preservation follows from the monotonicity of G and concatenation. Suppose the image of $(u_0, \dots, u_r)$ embeds into the image of $(v_0, \dots, v_r)$.

No old downset contains P . Lemma 4.4 therefore applies to the r displayed universal separators. Their cofinal destinations are the corresponding r displayed target universal separators.

Fix $i < r$. When $i > 0$, cofinality at the preceding separator forces all of $G(u_i)$ to map after the corresponding preceding target separator. Choose an image point η of the following source B_P inside the corresponding following target B_P ; such a point exists by cofinal localization. Every image point of $G(u_i)$ is before η . By Lemma 6.1, $G(u_i)$ has nonzero limit length. Applying Lemma 6.2 shows that none of its positions can be in this following target B_P . Thus the whole image of $G(u_i)$ lies in the intervening target component $G(v_i)$. For $i = 0$ the same argument applies without a preceding boundary. We have proved

$$G(u_i) \preceq G(v_i) \quad (i < r).$$

Reflection for G yields $u_i \preceq v_i$ for all $i < r$.

After the last source universal separator, cofinality forces the final source component u_r into the final target component v_r . Hence $u_r \preceq v_r$, completing product reflection. $\square$

Corollary 6.4 (One universal marker raises the level when padding exists). *Under the hypotheses of Theorem 6.3, if $o(X) = H_N$ and $n + |\mathcal{G}| = N$, then $o(Y) = H_{N+1}$.*

Proof. Apply the finite Cartesian-product theorem to the embeddings Ψ_r , take the supremum in (2.3), and combine with the atom-count upper bound, exactly as in Corollary 5.3. $\square$

Notice the role of the old proper block. It need not encode the whole component by itself. Its job is to give a limit-length ending, while an excluded finite letter makes that ending cancellable. These two properties are logically distinct and both are used.

6.3 The two constructions as one statement

Theorems 5.2 and 6.3 differ only in which guard map is attached to a component. It is convenient, and it is what the quick reference of Appendix A states, to record them as a single theorem.

Theorem 6.5 (Unified guarded product embedding). *Let D be a nonempty downset not in $\mathcal{G}$ with $D \not\subseteq E$ for every $E \in \mathcal{G}$, and assume either*

- (i) $D \neq P$, or
- (ii) $D = P$ and $\mathcal{G}$ contains a nonempty proper downset.

Define a guard map g_D on $\mathcal{W}(P, \mathcal{G})$ by

$$g_D(u) = \begin{cases} uc, & D \neq P, c \in P \setminus D, \\ ucB_R, & D = P, R \in \mathcal{G} \text{ proper}, c \in P \setminus R, \end{cases} \quad (6.4)$$

and put, for $m \geq 0$,

$$\Phi_m(u_0, \dots, u_m) = g_D(u_0)B_D g_D(u_1)B_D \cdots B_D g_D(u_m). \quad (6.5)$$

Then

$$\Phi_m(u_0, \dots, u_m) \preceq \Phi_m(v_0, \dots, v_m) \iff u_i \preceq v_i \text{ for every } i \leq m,$$

so Φ_m is an order embedding $\mathcal{W}(P, \mathcal{G})^{m+1} \hookrightarrow \mathcal{W}(P, \mathcal{G} \cup \{D\})$.

Proof. Both guard maps are order preserving and order reflecting and send $\mathcal{W}(P, \mathcal{G})$ into itself: for case (i) this is Lemma 5.1 together with the concatenation lemma, and for case (ii) it is Lemma 6.1. Preservation of Φ_m follows by concatenating componentwise embeddings with identity maps on the separators.

For reflection, Lemma 4.4 aligns the m displayed separators and Corollary 4.5 places each guarded component between the corresponding target boundaries. In case (i) the terminal letter c of $g_D(u_i)$ cannot lie inside the next explicit target separator, because $c \notin D$ and D is downward closed, and it cannot lie after that separator by Corollary 4.5(b); so the whole of $g_D(u_i)$ lies in $g_D(v_i)$. In case (ii) the word $g_D(u_i)$ has nonzero limit length by Lemma 6.1, so Lemma 6.2 forbids it from meeting the next explicit target separator at all, with the same conclusion. The last component is handled by Corollary 4.5(c). Reflection of g_D then gives $u_i \preccurlyeq v_i$ for every i . $\square$

Remark 6.6 (Guarding the final component or not). Theorem 6.5 guards *every* component, including the last one, whereas the maps Φ_r of (5.1) and Ψ_r of (6.3) leave the final component u_r unguarded. Both conventions are correct and both are used in the literature on this construction. With a trailing guard the final component is handled by cancellation, exactly like the others; without one it is handled by Corollary 4.5(c), since nothing follows it in the target. We take the unguarded form as the official construction of Sections 5 and 6, because it is the shorter expression, and record the all-guarded variant here and in Appendix A so that the two displays are not mistaken for a discrepancy.

6.4 An elementary alternative to the product theorem

Corollaries 5.3 and 6.4 pass from the Cartesian-power embeddings to the ordinal lower bound through Theorem 2.2(iii), the de Jongh–Parikh natural-product formula. That import can be avoided. The following lemma supplies, by hand, the only consequence of it that the main theorem needs, and the route is recorded here in full because it changes the hypotheses of the main theorem rather than merely restating it.

Lemma 6.7 (Finite powers amplify a maximal order type). *For every well-quasi-order X and every finite $r \geq 1$,*

$$o(X^r) \geq o(X)^r, \quad (6.6)$$

with ordinary ordinal exponentiation on the right.

Proof. Let L be a maximal linear extension of the quotient of X , of type $\alpha = o(X)$. Order r -tuples by comparing their *last unequal coordinate*, using L in that coordinate. This is a linear order, and it is a well-order of type α^r : the last coordinate indexes α consecutive copies of the corresponding order on $(r-1)$ -tuples, so the type satisfies $\alpha_r = \alpha_{r-1} \cdot \alpha$ with $\alpha_1 = \alpha$.

It extends the componentwise order. Indeed, if one tuple is componentwise below another and the two differ, then at every coordinate the first is below or equal in L , and at the last coordinate where they differ it is strictly earlier. Hence the last-unequal-coordinate order is a linear extension of X^r , and (6.6) follows from the definition of o . $\square$

Corollary 6.8 (One new block raises one Higman level, elementarily). *Let $X = \mathcal{W}(P, \mathcal{G})$ and $Y = \mathcal{W}(P, \mathcal{G} \cup \{D\})$ be as in Theorem 6.5, and suppose $o(X) = H_N$ with $n + |\mathcal{G}| = N$. Then $o(Y) = H_{N+1}$.*

Proof. For every $r \geq 1$, Theorem 6.5 gives an order embedding $X^r \hookrightarrow Y$, so $o(Y) \geq o(X^r) \geq H_N^r$ by Theorem 2.2(i) and Lemma 6.7. Taking the supremum over r and applying the ordinary-exponentiation identity (2.4) gives $o(Y) \geq H_{N+1}$. Proposition 3.8, with one more atom, gives the reverse inequality. $\square$

Only an explicit linear extension of a Cartesian product was used here. In particular, the argument does not require any general formula for the maximal order type of a product, nor any formula for lexicographic products of arbitrary partial orders. This is what the alternative route buys: with Corollary 6.8 in place of Corollaries 5.3 and 6.4, the proof of Theorem 1.1 uses from Theorem 2.2 only the monotonicity assertions (i) and (ii) and the finite-word formula (v), and in particular no natural-product theorem. The main line of the article nevertheless keeps the classical route, both because it is the shorter derivation and because assertions (iii) and (iv) are genuinely needed later, in the exact-length calculations of Section 10, which the elementary bound (6.6) does not supply.

7 The full classification and the main theorem

Theorem 7.1 (Selected-tail classification). *Let P be a nonempty finite poset of size n , and let $\mathcal{F} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$.*

(i) *If $\mathcal{F} \neq \{P\}$, then*

$$o(\mathcal{W}(P, \mathcal{F})) = H_{n+|\mathcal{F}|} = \omega^{\omega^{n+|\mathcal{F}|-1}}. \quad (7.1)$$

This includes the empty family.

(ii) *If $\mathcal{F} = \{P\}$, then*

$$o(\mathcal{W}(P, \{P\})) = H_n \cdot \omega = \omega^{\omega^{n-1}+1}. \quad (7.2)$$

Proof of part (i). For the empty family, (7.1) is the finite-word formula. Otherwise $\mathcal{F} \neq \{P\}$ implies that the family contains at least one proper downset. List its proper downsets as $D_1, \dots, D_t$ in increasing cardinality, breaking ties arbitrarily. If $P \in \mathcal{F}$, append it last.

Start with $X_0 = P^*$, of type H_n . At step $i \leq t$, no earlier downset contains D_i . Indeed, an earlier downset has no greater cardinality, and equality together with containment would make the downsets equal. Theorem 5.2 applies. Inductively, Corollary 5.3, or equally Corollary 6.8 if one prefers the elementary route, gives

$$o(\mathcal{W}(P, \{D_1, \dots, D_i\})) = H_{n+i}.$$

If P is absent from $\mathcal{F}$, this finishes the proof. If P is present, there is an old proper downset because $t \geq 1$. Apply Theorem 6.3 and Corollary 6.4 to the final step, obtaining $H_{n+t+1} = H_{n+|\mathcal{F}|}$. $\square$

Proof of part (ii). Every expression using only the universal infinite block can be written

$$u_0 B_P u_1 B_P \cdots u_{k-1} B_P v, \quad u_i, v \in P^*.$$

By (6.1), it is equivalent to $B_P^k v$. Here B_P^k means k successive copies, not an infinite repetition. These normal forms satisfy

$$B_P^k v \preccurlyeq B_P^\ell w \iff (k < \ell) \text{ or } (k = \ell \text{ and } v \preccurlyeq w). \quad (7.3)$$

If $k < \ell$, send the k source infinite blocks into the first k target blocks and the finite suffix into one extra universal target block. If $k > \ell$, k successive source infinite blocks require k distinct cofinal target infinite destinations by Lemma 4.1, which is impossible. If $k = \ell$, synchronization leaves the source suffix wholly in the target suffix, so $v \preccurlyeq w$ is necessary. It is plainly sufficient.

Thus the quotient is the ordered sum of ω copies of P^* :

$$\mathcal{W}(P, \{P\}) / \simeq \cong \sum_{k < \omega} P^*. \quad (7.4)$$

Every linear extension must put all of the k th copy before the $(k+1)$ st copy. Each copy has maximal order type H_n , and one may use a maximal linear extension in every copy. Therefore the maximal order type is the ordinary ordered sum $\sum_{k < \omega} H_n = H_n \cdot \omega$. The elementary identity $\omega^\alpha \cdot \omega = \omega^{\alpha+1}$ gives (7.2). $\square$

Proof of Theorem 1.1. Suppose $n \geq 2$. Choose a minimal element a of P . The singleton $\{a\}$ is a nonempty proper downset. Consequently the full family $\mathcal{J}(P) \setminus \{\emptyset\}$ is not the exceptional family $\{P\}$. Its size is $j(P) - 1$. Proposition 3.5 and Theorem 7.1(i) give

$$o(s_{\omega^2}^F(P)) = H_{n+j(P)-1} = \omega^{\omega^{n+j(P)-2}}.$$

For $n = 1$, there is one word at each ordinal length $\alpha < \omega^2$, and embedding is exactly comparison of lengths. Its quotient is therefore the ordinal ω^2 itself. Equivalently, apply part (ii) with $H_1 = \omega$. For $n = 0$, the empty word is the sole word, giving maximal order type 1. $\square$

7.1 Why the supremum argument is legitimate

The proof never identifies the order type of an increasing union with the supremum of the order types of its pieces. Such an identification is not a general principle for maximal order types. Instead, for each fixed finite r , there is a separately verified order embedding of a Cartesian power. It gives a genuine lower bound on the *same* target order type. Taking the supremum of a collection of lower bounds is valid. The upper bound is independently obtained by a monotone surjection.

The distinction is not pedantic, and Section 10 makes it concrete. There the maximal order type of each individual length stratum is computed exactly, and the supremum of those values, recorded in (10.7), is ω^{ω^n} : strictly smaller than the answer $\omega^{\omega^{n+j(P)-2}}$ of Theorem 1.1 whenever $n \geq 2$. A stratum-by-stratum supremum therefore genuinely fails to compute the theorem, while the supremum of lower bounds used here does not.

7.2 The singleton warning is structural

For a singleton alphabet, two formal atoms are available: the letter and its periodic ω -block. Their finite words would have maximal type $H_2 = \omega^\omega$ if treated as a free Higman alphabet. But after ordinal evaluation, the actual type is only ω^2 . The family $\mathcal{F} = \{P\}$ exhibits the same defect over every nonempty finite alphabet. Hence atom counting alone cannot be a lower-bound proof. The padding construction explains exactly why the full family over $n \geq 2$ avoids this collapse.

8 The binary proof written out explicitly

This section specializes the general proof so the selected problem can be checked without tracking an arbitrary downset family.

8.1 Two incomparable letters

Let $P = \{a, b\}$ with no comparisons between distinct letters, and abbreviate

$$A = a^\omega, \quad B = b^\omega, \quad U = (ab)^\omega.$$

There are exactly three nonempty downsets, giving these three periodic block types. Define stages by their allowed tokens, and record the guard used to activate the block that is new in each row:

stage	allowed tokens	guard for the new block	claimed maximal type
X_0	a, b	none; finite words	$H_2 = \omega^\omega$
X_1	a, b, A	$u \mapsto ub$	$H_3 = \omega^{\omega^2}$
X_2	a, b, A, B	$u \mapsto ua$	$H_4 = \omega^{\omega^3}$
X_3	a, b, A, B, U	$u \mapsto ub a^\omega$	$H_5 = \omega^{\omega^4}$

The upper bound in each row comes from the number of tokens. The guard in the last row is the limit padding G of Lemma 6.1; the guards in the two middle rows are single excluded letters. The first equality is classical. To pass from X_0 to X_1 , use, for every finite r ,

$$(u_0, \dots, u_r) \mapsto u_0 b A u_1 b A \cdots u_{r-1} b A u_r.$$

Source A -blocks must synchronize with target A -blocks, and the guard b cannot enter A . This embeds X_0^{r+1} .

To pass from X_1 to X_2 , use

$$(u_0, \dots, u_r) \mapsto u_0 a B u_1 a B \cdots u_{r-1} a B u_r.$$

A source B -block cannot be cofinal in an old target A -block, and its preceding guard a cannot enter a target B -block. This embeds X_1^{r+1} .

For the final step, set

$$G(u) = ubA.$$

The guard b cannot enter the final A , so G reflects embedding; the image has limit length. Now use

$$(u_0, \dots, u_r) \mapsto G(u_0)UG(u_1)U \cdots G(u_{r-1})Uu_r.$$

A source U contains both letters infinitely often, so neither A nor B can be its cofinal destination. The displayed U -blocks synchronize. Limit padding prevents the intervening components from entering the next U before a fixed image point of its source marker. This embeds X_2^{r+1} .

At each of the three steps, (2.3) converts these product embeddings into the next displayed ordinal. Finally, every binary word of length below ω^2 is equivalent to an expression in a, b, A, B, U . This proves the sharpness of the binary antichain bound.

8.2 Two ordered letters

Let $0 < 1$. The nonempty downsets are $\{0\}$ and $\{0, 1\}$; their periodic blocks may be represented by

$$A = 0^\omega, \quad U = 1^\omega.$$

The latter is equivalent to $(01)^\omega$. Begin with finite words over $\{0, 1\}$, of maximal type H_2 . Add A using the guard 1, giving H_3 . Then add U using the limit padding $G(u) = u1A$, giving H_4 . Thus the chain's sharpened upper bound is also attained.

8.3 Three easily confused comparisons

For the antichain, the following facts illustrate the mechanism:

$$aU \simeq U, \quad A \preccurlyeq U, \quad AB \not\preccurlyeq U.$$

The first is finite-prefix absorption. The second uses cofinal recurrence of a in U . The third follows because two successive ω -blocks cannot both have cofinal image in the single target block. Also $U \not\preccurlyeq AB$: its tail would have to lie in either the pure a block or the pure b block, losing one of its recurrent labels. These facts show why both order of occurrence and cofinal type matter.

8.4 Negative controls: three invalid shortcuts

The hypotheses of Theorems 5.2 and 6.3 are not decorative. Each of the following three simplifications is tempting, and each fails on the two-letter antichain $P = \{a, b\}$ with an explicit witness. The three witnesses are also executed as negative controls in the verification suite of Section 12, where they are required to *fail*, so a regression that silently weakened a hypothesis would be caught.

Shortcut 1: omitting the guards. For any letter a ,

$$aB_P \simeq B_P, \quad \text{while} \quad a \not\preccurlyeq \varepsilon. \quad (8.1)$$

A component placed directly before a separator can therefore disappear under embeddability, so an unguarded tuple map need not reflect the componentwise order. This is (6.1) read as a counterexample.

Shortcut 2: using a finite guard at the universal block. Appending one letter does not repair the previous failure:

$$aaB_P \preccurlyeq aB_P, \quad \text{while} \quad a \not\preccurlyeq \varepsilon. \quad (8.2)$$

More generally, any fixed finite padding is absorbed into a universal periodic tail by (6.1). The guard of Lemma 6.1 is of limit length for a mathematical reason, not for convenience.

Shortcut 3: inserting a downset after one that contains it. The newness condition (4.1) cannot be dropped. Let the old family be $\mathcal{G} = \{E\}$ with $E = P$, and let the new downset be $D = \{a\}$, which is contained in E ; take the naive terminal guard $b \notin D$. Consider the two component pairs

$$(\varepsilon, a) \quad \text{and} \quad (B_E, \varepsilon).$$

The first is *not* componentwise below the second, since $a \not\leq \varepsilon$. Nevertheless their guarded encodings satisfy

$$b a^\omega a b \preceq B_E b a^\omega b. \quad (8.3)$$

Indeed, match the initial source b inside B_E , then the source a^ω cofinally inside that same B_E ; then match the remaining source a into the displayed target a^ω and the final source b to the terminal target b . The source separator has used the *old containing* block rather than its designated target separator, and synchronization fails. Introducing downsets in nondecreasing cardinality, as in the proof of Theorem 7.1, is exactly what rules this out: a strictly earlier downset of no larger size cannot contain the new one.

The article asserts the necessity of $D \not\subseteq E$ in prose several times; (8.3) is the worked instance behind those assertions.

9 Consequences and finite-poset examples

9.1 Chains, antichains, and extremal values

Corollary 9.1. *For $n \geq 2$,*

$$o(s_{\omega^2}^F(C_n)) = \omega^{\omega^{2n-1}}, \quad o(s_{\omega^2}^F(A_n)) = \omega^{\omega^{n+2n-2}}.$$

Among n -element posets, the first value is the minimum, attained exactly by chains, and the second is the maximum, attained exactly by antichains.

Proof. A chain has exactly $n + 1$ downsets, its initial segments. Every subset of an antichain is a downset, so it has 2^n downsets. For any finite poset, the initial segments of a linear extension give $n + 1$ distinct downsets, and there are at most 2^n subsets in total.

For the equality statement at the lower end, fix a linear extension. If there are only $n + 1$ downsets, every downset is one of its initial segments. In particular all principal downsets are comparable by inclusion. Since $\downarrow x \subseteq \downarrow y$ implies $x \leq_P y$, every pair of elements is comparable, and P is a chain. At the upper end, if every subset is a downset, a strict relation $x <_P y$ would make $\{y\}$ fail to be a downset. Thus P is an antichain. The function of $j(P)$ in (1.1) is strictly increasing, so the ordinal extremal assertions follow. $\square$

9.2 Equivalent counting invariants

Every downset of a finite poset is generated by its maximal elements, which form an antichain. Conversely, the maximal elements of the downset generated by an antichain are that antichain. These maps give a bijection, including the empty set. Thus $j(P)$ may equally be described as the number of antichains of P .

Complementation gives a bijection between downsets and upsets. Passing to the dual poset exchanges downsets with upsets, so $j(P^{\text{op}}) = j(P)$. The formula therefore yields

$$o(s_{\omega^2}^F(P^{\text{op}})) = o(s_{\omega^2}^F(P)).$$

This is equality of maximal order types, not a claim that the word quasi-orders are isomorphic.

For a disjoint union $P \amalg Q$, choosing a downset is equivalent to choosing one downset in each component, hence $j(P \amalg Q) = j(P)j(Q)$. For an ordered sum $P + Q$, in which every element of P lies below every element of Q , one has $j(P + Q) = j(P) + j(Q) - 1$. The two possibilities are a downset contained in P , or all of P together with a nonempty downset of Q . These identities make (1.1) explicit for many recursively assembled alphabets.

Alphabet P	n	$j(P)$	$\rho(P)$	Maximal order type
Two-element chain	2	3	3	ω^{ω^3}
Two-element antichain	2	4	4	ω^{ω^4}
Three-element chain	3	4	5	ω^{ω^5}
Three-element V or its dual	3	5	6	ω^{ω^6}
Two-element chain plus an isolated point	3	6	7	ω^{ω^7}
Three-element antichain	3	8	9	ω^{ω^9}
Four-element chain	4	5	7	ω^{ω^7}
Four-element diamond	4	6	8	ω^{ω^8}
Four-element antichain	4	16	18	$\omega^{\omega^{18}}$
Five-element antichain	5	32	35	$\omega^{\omega^{35}}$

Table 1: Exact consequences of Theorem 1.1. The diamond has one bottom element, two incomparable middle elements, and one top element.

9.3 A table of examples

In Table 1, $\rho(P) = n + j(P) - 2$ is the finite exponent parameter in $\omega^{\omega^{\rho(P)}}$.

The alphabet cardinality by itself is not enough: the two binary values already differ. Conversely, nonisomorphic alphabets can have the same maximal order type. For instance the three-element V and its dual have the same n and j , while a four-element chain and a three-element chain plus an isolated point have the same value of $n + j - 2$ despite different cardinalities.

9.4 Finite quasi-orders

For a finite quasi-order rather than a poset, first pass to its quotient by mutual comparison. Projection of labels preserves and reflects word embedding: a comparison of quotient labels is exactly the comparison of any representatives. Therefore the theorem applies with n equal to the number of equivalence classes and j equal to the number of downsets of the quotient poset. Counting raw labels instead of classes can give an incorrect answer.

10 Exact-length strata

This section computes the maximal order type of each individual length stratum. The results are supplementary: they are not used anywhere in the proof of Theorem 1.1, and a reader checking only the resolution of the two-letter question may skip the section entirely. Their point is negative, and it is made in Section 10.3: the answer of Theorem 1.1 cannot be assembled stratum by stratum.

Unlike the main line, this section does use the natural-product and natural-sum identities of Theorem 2.2(iii) and (iv). The elementary substitute of Lemma 6.7 is a one-sided bound and does not suffice here.

Throughout, P is a nonempty finite poset with $n = |P|$, and

$$\mathcal{E}_\alpha(P) = \{u : \text{len}(u) = \alpha\}, \quad d = j(P) - 2$$

is the number of *proper* nonempty downsets of P . Thus $d \geq 1$ exactly when $n \geq 2$, and $d = 0$ for a singleton alphabet.

10.1 A single infinite block

Proposition 10.1 (Length exactly ω). *For every nonempty finite poset P ,*

$$o(\mathcal{E}_\omega(P)) = H_n \cdot d + 1, \quad (10.1)$$

$$o(s_{\omega+1}(P)) = H_n \cdot (d + 1) + 1, \quad (10.2)$$

where $s_{\omega+1}(P)$ denotes the words of length at most ω .

Proof. By Lemma 3.4, every word of length ω is equivalent to pB_D with $D = I(u)$ and p empty or ending outside D . Group these classes into *fibres* indexed by the nonempty downset D .

Fix a proper nonempty D and consider its fibre. For two normalized forms,

$$pB_D \preccurlyeq qB_D \iff p \preccurlyeq q. \quad (10.3)$$

The implication from right to left is monotonicity of concatenation. For the other direction, the claim is trivial when $p = \varepsilon$; otherwise the final letter of p lies outside D , so by Lemma 3.1 it cannot be matched inside the target periodic tail, and its image lies in the target prefix q . Every earlier position of p maps strictly before it, hence also into q .

By (10.3) the map $p \mapsto pB_D$ is an isomorphism from the suborder of P^* consisting of the admissible prefixes, that is the words that are empty or end outside D , onto the fibre of D . Hence that fibre has type at most $o(P^*) = H_n$. In the other direction choose $c \notin D$ and map $u \in P^*$ to ucB_D , whose prefix uc is admissible. This map is order preserving, and order reflecting by (10.3) together with Lemma 5.1. Hence the fibre of D has type exactly H_n .

The fibre of the full downset P is a single equivalence class: by (6.1) every pB_P is equivalent to B_P . That class lies above every word of length ω , since $D \subseteq P$ always holds. Conversely, by Lemma 4.1, an embedding between words of length ω forces inclusion of their recurrent downsets, so comparisons across fibres go only from smaller to larger downsets.

For the lower bound, order the d proper nonempty downsets by nondecreasing cardinality, concatenate maximal linear extensions of their fibres in that order, and place the single class B_P last. No comparison is violated, because a comparison between distinct fibres always runs from a downset of strictly smaller cardinality to a larger one. This is a linear extension of type $H_n \cdot d + 1$.

For the upper bound, erase every comparison between distinct fibres. The result is a disjoint union of d orders of type H_n and one singleton, whose maximal order type is $H_n \oplus \dots \oplus H_n \oplus 1 = H_n \cdot d + 1$ by Theorem 2.2(iv); the natural sum of d copies of $\omega^{\omega^{n-1}}$ with 1 collects the coefficients at one exponent and leaves the finite part alone. Deleting comparisons cannot decrease the maximal order type, so the original order has type at most this value. This proves (10.1).

For (10.2), adjoin the finite words. No infinite word embeds into a finite word, so placing all of P^* first respects every comparison and adds one further layer of type H_n ; erasing the comparisons between P^* and $\mathcal{E}_\omega(P)$ gives the matching upper bound, again by Theorem 2.2(iv). $\square$

10.2 Several blocks and a fixed finite tail

Lemma 10.2 (Rigidity at equal block count). *Let u and v each have exactly r canonical ω -blocks, and suppose $u \preccurlyeq v$. Then the i th canonical source block is mapped, entirely and cofinally, inside the i th canonical target block, for every $i < r$, and the finite source tail is mapped inside the finite target tail.*

Proof. By Lemma 4.1, applied to the periodic normal form of each canonical source block, each source block has a cofinal destination among the target blocks, distinct source blocks have distinct destinations, and the assignment is increasing. An increasing injection of an r -element chain into itself is the identity, so the i th source block is cofinal in the i th target block.

The i th source block cannot have an image point before the end of the $(i - 1)$ st target block, because the $(i - 1)$ st source block has already used that block cofinally and every point of the i th source block comes later. It cannot have an image point after the end of the i th target block either: if it did, the infinitely many source positions following that one would map beyond the i th target block, leaving only finitely many image points inside it, contradicting cofinality there. Hence the whole i th source block lies inside the i th target block. After the last block only the finite target tail remains, which handles the tail. $\square$

Lemma 10.2 is strictly stronger than Lemma 4.1 with Remark 4.2, and it is not available in the main argument: it needs the hypothesis that source and target have the same number of canonical blocks, which is exactly what the product constructions of Sections 5 and 6 do not assume. Within the present section that hypothesis is automatic, because the length is fixed.

Proposition 10.3 (Fixed-length product decomposition). *For all $r, k < \omega$ there is an isomorphism of quotient orders*

$$\mathcal{E}_{\omega r+k}(P) \cong \mathcal{E}_\omega(P)^r \times P^k, \quad (10.4)$$

and consequently, writing $A = H_n \cdot d + 1$,

$$o(\mathcal{E}_{\omega r+k}(P)) = A^{\otimes r} \otimes n^k. \quad (10.5)$$

Proof. Concatenation is a bijection between r -tuples of length- ω words together with a length- k tail and words of length $\omega r + k$, since the canonical block decomposition of such a word is unique. By Lemma 10.2, an embedding between two such words restricts to an embedding of the i th block into the i th block and of the tail into the tail; conversely, componentwise embeddings concatenate by Lemma 2.1. A strictly increasing injection between finite chains of the same length k is the identity, so the induced relation on the tails is exactly the componentwise label order of P^k . This proves (10.4), and (10.5) follows from Theorem 2.2(iii), Proposition 10.1, and $o(P^k) = n^k$ for the finite poset P^k . $\square$

Because $H_n = \omega^{\omega^{n-1}}$, expanding the natural power in (10.5) gives the Cantor normal form

$$o(\mathcal{E}_{\omega r+k}(P)) = \sum_{i=r, r-1, \dots, 0} \omega^{\omega^{n-1} \cdot i} \left(n^k \binom{r}{i} d^i \right). \quad (10.6)$$

The sum is written in decreasing i , so it is an ordinary Cantor-normal-form sum, and terms with zero coefficient are omitted. The coefficient arises as follows: in the expansion of $(H_n \cdot d + 1)^{\otimes r}$ one chooses, from each of the r factors, either the leading monomial $H_n \cdot d$ or the constant 1; choosing the leading monomial in exactly i of the factors can be done in $\binom{r}{i}$ ways, contributes the finite coefficient d^i , and produces the exponent $\omega^{n-1} \oplus \dots \oplus \omega^{n-1}$ with i summands, namely $\omega^{n-1} \cdot i$. The convention $d^0 = 1$ is in force, including when $d = 0$: for a singleton alphabet (10.6) correctly collapses to $n^k = 1$.

10.3 A warning about taking suprema over lengths

Suppose $n \geq 2$, so that $d \geq 1$. Taking the supremum of (10.6) over all lengths gives

$$\sup_{r, k < \omega} o(\mathcal{E}_{\omega r+k}(P)) = \omega^{\omega^n}, \quad (10.7)$$

because the leading exponents are $\omega^{n-1} \cdot r$, whose supremum over r is ω^n , and finite coefficients and lower-order terms do not affect that supremum.

The value (10.7) is *strictly below* the answer $\omega^{\omega^{n+j(P)-2}}$ of Theorem 1.1. To see the strictness, take a linear extension of P : its $n+1$ prefixes are distinct downsets, so $j(P) \geq n+1$ and therefore $n+j(P)-2 \geq 2n-1 > n$ whenever $n \geq 2$.

There is no contradiction, and the explanation is the one already given in Section 7.1. The length strata are *not* ordinal-sum layers of the whole order. Ordinal length is indeed monotone under embeddability, by Lemma 4.1 for the block count and Lemma 10.2 for the tail. But an ordinal-sum layering would require every word of a shorter length to embed into every word of a longer one, and that fails at once: over the two-letter antichain, b has length 1 and a^ω has length ω , yet $b \not\leq a^\omega$. A maximal linear extension of the full order is therefore free to interleave strata and exploit incomparabilities between them, and it does. The downset induction of Section 7 detects that additional complexity; the separate exact-length calculations, by construction, cannot.

11 An exact symbolic embedding algorithm

11.1 Representation

The supplied module `code/omega2.py` represents a word by a finite sequence of tokens. A token $(\text{'f'}, x)$ is a finite letter, and $(\text{'w'}, D)$ is B_D , with D stored as a nonempty bitmask downset. The poset relation is supplied by predecessor masks. The code validates reflexivity, antisymmetry, and transitivity.

This representation does not claim to compute the recurrent set of an arbitrary black-box infinite input. Rather, it decides embeddings between already supplied finite periodic-block descriptions. The mathematical normal-form theorem says such descriptions represent every equivalence class in the target order.

11.2 Greedy decision procedure

Maintain a pointer to a target token, initially the first. Read source tokens from left to right.

Source token is a finite letter a .

Scan target tokens until finding either a finite letter b with $a \leq_P b$, or an infinite token B_E with $a \in E$. In the finite case consume the letter and advance the pointer. In the infinite case leave the pointer at B_E : after finitely many matches, an equivalent periodic tail remains.

Source token is an infinite block B_D .

Scan target tokens until finding an infinite token B_E with $D \subseteq E$. Consume its tail cofinally and advance the pointer to the following token.

If the required token does not exist, reject. Accept after every source token has been processed.

The same rules, written as a table of the four possible source/target token pairs, are the following. A letter token is written a or b and an infinite token is written $[D]$ or $[E]$, denoting B_D and B_E .

Source	Target	Matching condition	Cursor after a match
a	b	$a \leq_P b$	Advance past b
a	$[E]$	$a \in E$	Remain at $[E]$
$[D]$	$[E]$	$D \subseteq E$	Advance past $[E]$
$[D]$	b	Never	No match

When the current target token cannot match, skip it and advance the cursor. The last row is worth stating explicitly: an infinite source token never matches a finite target token, since a strictly increasing map out of ω cannot land in a single position. The second row is the only one that leaves the cursor where it is, because a finite match consumes only finitely many positions of a periodic tail and leaves an equivalent tail behind.

Theorem 11.1 (Exactness of the algorithm). *The greedy procedure accepts exactly when the represented source ordinal word embeds into the represented target ordinal word.*

Proof. Within a periodic target block, removing any finite initial segment leaves an equivalent tail. Thus the symbolic state need not record a phase or a finite offset in that block.

For a finite source letter, the first eligible target token provides an early possible match. Choosing a later token cannot leave a more useful suffix. Within an infinite token, choosing an actual sufficiently early eligible occurrence leaves an equivalent periodic tail. Therefore a successful continuation after any possible match can also be made after the greedy match.

For an infinite source block B_D , Lemma 4.1 shows that every embedding must use some target block B_E cofinally with $D \subseteq E$. All later source positions must then lie after that target block. The greedy choice takes the first eligible such block. The entire source B_D embeds in its available tail by Lemma 3.1; allowing part of the source block to occur earlier cannot improve the endpoint beyond this first eligible block. The remaining target suffix after the greedy choice contains the suffix left by any later choice.

Inducting on the number of remaining source tokens proves completeness: whenever a completion exists, the greedy step preserves the existence of a completion. Conversely, each accepted finite step can

be realized by an actual letter match in a finite token or a periodic tail, and each accepted infinite step by a cofinal embedding into the selected target block. Concatenating these realizations gives an ordinal embedding, so acceptance is sound. $\square$

The target pointer never retreats. With bit operations and relation lookup counted as unit cost, the decision procedure uses $O(|u|_{\text{tok}} + |v|_{\text{tok}})$ time and constant auxiliary space. For unbounded alphabet size, bitmask operations have their ordinary bit costs; validating masks and the poset incurs additional cost. The reference checker uses memoized exploration of every eligible target endpoint rather than the first endpoint.

11.3 Canonical quotient representatives

Proposition 3.5 produces finite expressions but, without its normalization clause, not unique ones. The normalization is easy to carry out on token expressions, and it makes the representatives canonical.

The normalization procedure. Process the tokens from left to right onto a stack. A letter token is pushed. When an ideal token $[D]$ arrives, first pop the immediately preceding letter tokens for as long as their letters belong to D , and then push $[D]$. The final finite tail, that is the letters following the last ideal token, is never altered. Equivalently: in each finite prefix preceding an ideal token $[D]$, delete the longest suffix all of whose letters lie in D .

Each deleted suffix is absorbed by the ideal token that follows it, by (3.2), so the output is equivalent to the input; and the output is exactly an expression of the shape (3.4), since every prefix is now empty or ends outside the downset of the block that follows it. The procedure is linear in the number of tokens and is implemented as `normalize` in `code/omega2.py`.

Proposition 11.2 (Uniqueness of normalized expressions). *Two normalized token expressions over a finite poset P are mutually embeddable if and only if they are identical.*

Proof. One direction is trivial. For the other, suppose u and v are normalized and mutually embeddable. Ordinal length is monotone under embeddability, so u and v have equal lengths; hence they have the same number r of canonical ω -blocks and finite tails of the same length.

By Lemma 10.2, applied in both directions, the i th block of u and the i th block of v are mutually embeddable, and so are the two finite tails. Mutual embeddability of blocks forces mutual inclusion of their downsets by Lemma 3.1, so the i th ideal tokens agree, say both equal to $[D_i]$. Since both expressions are normalized, the finite prefixes p_i and q_i preceding these tokens are empty or end outside D_i , so (10.3) turns the mutual embeddability of the blocks into mutual embeddability of p_i and q_i .

Finally, two finite words over a poset that embed into each other have equal finite length, so both index maps are the identity; antisymmetry of $\leq_P$ then forces equality of corresponding labels. Hence $p_i = q_i$ for every i , the tails agree, and $u = v$. $\square$

Consequently the quotient of $s_{\omega^2}^F(P)$ has a countable, effectively comparable system of canonical finite representatives, even though there are uncountably many literal infinite words over an alphabet with at least two elements. The forward reference promised after Proposition 3.5 is discharged here.

11.4 Why finite expansion is not an adequate substitute

Replacing each B_D by M repetitions for a large integer M changes the problem. The absorption equivalence $aB_{\downarrow a} \simeq B_{\downarrow a}$ already fails for the direction from length $M+1$ to length M after a uniform finite expansion. Conversely, finite approximations do not directly certify the cofinal-boundary restrictions needed for reflection of product maps. The supplied code therefore treats ω -blocks symbolically and uses the proven cofinal semantics exactly.

12 Computational checks and reproducibility

The included verification run passed 3,240,040 assertions. Its random seed is 20260919; all random checks supplement deterministic checks. The report is stored in `data/verification.json`. The recorded run

used Python 3.14.4, and the code targets Python 3.10 or newer without third-party packages. The suite is a single program: the counts below come from one run, and they are not the sum of two separately reported runs.

Check	Number of assertions
Exhaustive greedy/reference agreement, small posets	38,107
Exhaustive greedy/reference agreement, binary domain	726,242
Seeded longer-word greedy/reference agreement	9,800
Normalization preserves every decided relation	726,242
Normalization preserves the class and the ordinal length	2,244
Mutually embeddable pairs have identical normal forms	4,428
Exhaustive proper/universal product maps	1,544,960
Binary support-family product maps	44,667
Seeded longer-word product maps	35,500
Padding reflection	55,150
Padding has nonzero limit length	4,024
Universal-only comparison formula	20,640
Universal-only normalization	2,660
Fixed-length product decomposition	25,000
Maximal elements generate their downset	356
Boundary, absorption, and regression cases	16
Negative controls, required to fail	4
Total	3,240,040

Table 2: Executed symbolic checks. A “word length” in the test generator means token length, not ordinal length.

The exhaustive comparison of the two deciders uses all naturally labelled posets of sizes zero through three and every source/target token word of length at most two. The longer-word comparisons cover all naturally labelled posets of sizes two through four and use token words of length at most twelve.

12.1 The exhaustive binary domain

Over the two-letter alphabets the whole token domain up to length four is enumerated and every ordered pair is tested. The antichain has five tokens, giving $1 + 5 + 5^2 + 5^3 + 5^4 = 781$ expressions and $781^2 = 609,961$ ordered pairs; the chain has four tokens, giving 341 expressions and 116,281 pairs, for 726,242 ordered pairs in all. For each pair the greedy and reference comparators agree, normalization preserves the decided relation, and mutually embeddable expressions have identical normal forms, which is the executable form of Proposition 11.2.

Across the same two alphabets, every allowed support subfamily is used as an insertion chain, giving 14 activation stages; the universal-only exceptional activation is excluded, as the hypotheses require. Support prefixes shared by different subfamilies are deliberately retested.

The three invalid shortcuts of Section 8.4 are executed as negative controls: the suite asserts that the naive encodings do *not* reflect the componentwise order, so weakening a hypothesis would be detected rather than silently accepted. The fixed-length decomposition of Proposition 10.3 is checked on randomly generated words of prescribed length $\omega r + k$ over every naturally labelled poset of size at most four.

For the product embeddings, every naturally labelled poset of sizes two through four is processed, introducing all nonempty downsets in cardinality order. There are 355 marker stages. At each stage the exhaustive test uses a fixed four-word component pool and all pairs of tuples of arities two and three. The seeded tests add tuples of arities two through five with component token lengths at most eight. In total these give 1,625,127 product-map comparisons. Separate padding tests allow all one-token words, including already universal words, so padding reflection is checked beyond the narrower family needed in the main induction.

The two embedding implementations share the mathematical periodic-tail semantics. Agreement between them is a check of the greedy optimization, not an independent derivation of that semantics. Likewise, no collection of the finite assertions in Table 2 proves ω^{ω^4} to be the maximal order type. That conclusion rests on the unbounded family of product embeddings and the ordinal calculation.

12.2 Enumeration conventions

The auxiliary enumerator considers relations on $\{0, \dots, n-1\}$ whose strict comparisons respect the usual numerical order. Every finite-poset isomorphism class has such a labelling, but the enumeration is neither an enumeration of all labellings nor an isomorphism-free enumeration. The numbers of generated relations for $n = 0, 1, 2, 3, 4, 5$ are respectively

$$1, 1, 2, 7, 40, 357.$$

These counts were computed by the included program. Histograms by the number of downsets appear in `data/ideal_counts.csv`. The 50 individual posets of sizes one through four, with their predecessor and successor masks, downset counts, and the exponent predicted by (1.1), are listed in `data/finite_posets.csv`; the row for $n = 1$ carries the literal entry `exception: omega^2` rather than a numerical exponent. Table examples are also exported to `data/examples.csv`. No claim about uniform sampling of isomorphism classes is intended.

To compute $j(P)$ for one small input poset, the code simply checks all 2^n subsets for downward closure. This exponential enumeration is sufficient for the examples and the proof tests. The article does not claim a polynomial-time algorithm for downset counting.

12.3 Reproduce the package

From the root of the unpacked directory:

```
python3 code/verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Python command regenerates the verification reports and CSV tables. It accepts `-seed`, `-samples`, `-pair-bound`, and `-output-dir`; the defaults are 20260919, 250, 4, and `data/`. Raising `-pair-bound` enlarges the exhaustive binary enumeration very quickly. A three-pass `pdflatex` build is also supplied as `build.sh`. The PDF can alternatively be built by running `pdflatex article.tex` repeatedly until cross-references stabilize. No external bibliography processor is required; a separate BibTeX file is supplied for reuse.

13 Proof audit, scope, and further directions

13.1 Where a hidden gap would have to occur

The proof is organized so its nonclassical burden is concentrated in a few claims. The atom-count upper bound is a monotone-surjection argument. Cofinal localization uses finiteness of the target token decomposition. Synchronization uses the newness condition $D \not\subseteq E$ for all old blocks, not merely distinctness of block labels. The proper-marker reflection uses downward closure to exclude a guard from the next marker. The full-marker reflection uses *both* cancellation and the limit length of the padded component.

These dependencies explain several tempting but incorrect shortcuts. A source infinite block need not land wholly inside one target token. The ordinary evaluation map from atom strings need not reflect order. The universal marker admits no excluded finite letter. A finite test cannot certify an infinite supremum. None of these shortcuts is used.

The four boundaries to check independently. Concretely, four points carry the argument, and a hidden gap would have to sit at one of them.

- (1) Embeddings are *strictly* increasing on ordinal positions. Without strictness, cofinal consumption of a target block fails and Lemma 4.1 is false as stated.
- (2) A newly introduced downset is contained in *no* old downset, condition (4.1). Mere distinctness is not enough, and (8.3) is the explicit counterexample.
- (3) Exclusion applies to the *explicit pure periodic segment* of a displayed separator, not to the finite prefix that may precede it inside the same canonical block. This is Remark 4.3.
- (4) Full support requires an *earlier proper* support, so that a guard of positive limit length can be built. Without one, the exceptional value (7.2) is the truth.

The negative controls of Section 8.4 exhibit the failure of the second and fourth safeguards, and of guarding altogether; the singleton alphabet illustrates the resulting exceptional maximal order type.

A reader's guide. An independent referee can focus first on Theorems 5.2 and 6.3, or equivalently on their unified form Theorem 6.5. Once their product-reflection assertions are accepted, the classification follows by a short finite induction and the classical finite-product formula, or by the elementary Corollary 6.8 if one prefers to avoid that import. The binary specialization in Section 8 provides a second presentation of the same core argument, not an independent proof with different foundations.

Section 10 may be omitted entirely by a reader checking only the resolution of the original two-letter question. It is the one part of the article that uses the natural-product and natural-sum identities Theorem 2.2(iii) and (iv) essentially, and nothing in Sections 1–9 depends on it. Its role is to document, in (10.7), why the answer cannot be obtained one length at a time.

13.2 Established within the argument versus not claimed

Within this manuscript, the main theorem has a complete proof relative to the classical results stated in Theorem 2.2. There is no remaining conjectural lemma inside that derivation. Nevertheless the argument has not undergone independent expert review or proof-assistant verification. The appropriate external status is therefore a *proposed solution*, with explicit hypotheses and reproducible supporting checks, rather than a certified resolution.

The article does not compute all invariants of these quasi-orders: their height, width, and exact isomorphism classification are different tasks. It does not show that two alphabets with equal $n + j(P)$ produce isomorphic word orders. It does not give a canonical representation for arbitrary infinite input streams. It also does not assert that selected block families at higher ordinal cutoffs have the same counting law.

13.3 Why the proof stops at this cutoff

The decisive compactness feature is that a target word below ω^2 has only finitely many ω -blocks. The image of a source ω -block must eventually stabilize in one of them. At length ω^2 , a source ω -block may instead visit infinitely many successive target ω -blocks, taking only finitely many points from each. Thus the cofinal localization lemma in its present form is no longer valid when that target length is allowed.

A higher-cutoff extension would need a hierarchy of separator ranks and a new localization theorem that accounts for escape through infinitely many lower-rank blocks. Counting indecomposable blocks without that theorem would reproduce the same logical mistake that naive atom counting makes at the universal-only stage.

For infinite alphabets, even when every individual word has finite image, the set of possible recurrent downsets need not be finite, and a single finite inclusion ordering does not exhaust the construction. The present finite induction therefore does not directly transfer. These observations identify precise missing mechanisms for extensions; they are not reasons to reject further investigation.

13.4 A concrete formal-verification target

A useful formalization would first define ordinal concatenation and its embedding relation for finite periodic-token codes, then prove the exact symbolic decision theorem. The next targets are the cofinal

boundary lemma, proper-marker reflection, and universal-marker reflection. With the classical maximal-order-type product and finite-word theorems available, the final induction has very little additional complexity. The package contains executable tests but no Lean development, and it makes no claim of machine-checked proof.

14 Conclusion

The finite generator upper bound for words below ω^2 is sharp for every finite poset alphabet with at least two elements. The quantity that governs the answer is the number of letters plus the number of nonempty downsets, and not the alphabet's cardinality alone: the two-letter chain and the two-letter antichain already separate.

The mechanism behind the sharpness is explicit. Each proper nonempty downset can be activated as a separator by a terminal-letter guard, because a letter outside a downward-closed set cannot be absorbed by the block that repeats it. The full downset admits no such letter and can be activated only after an earlier proper downset has supplied a guard of positive limit length, which no finite initial segment of a periodic block can accommodate. When no earlier proper downset is available the activation genuinely fails, and the universal-only family has maximal order type $H_n \cdot \omega$ rather than H_{n+1} ; the singleton alphabet is the same phenomenon.

The proposed solution is therefore not merely a numerical lower bound matching a known upper bound. It identifies and isolates the structural mechanism responsible for the sharpness, in a form that is stated for arbitrary finite posets, classified for arbitrary allowed families of periodic tails, made effective by a canonical normal form and a linear-time symbolic comparator, and audited both by worked negative controls and by a reproducible test suite. Its external status remains that of an unrefereed proposed solution, in the sense set out in Section 13.

A A compact self-contained statement of the construction

This appendix restates the lower-bound mechanism on one page, in the all-guarded form of Theorem 6.5 and Remark 6.6. Fix a nonempty finite poset P , an allowed family $\mathcal{G} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$, and a new nonempty downset $D \notin \mathcal{G}$ with $D \not\subseteq E$ for every $E \in \mathcal{G}$. Set

$$g_D(u) = \begin{cases} uc, & D \neq P, \quad c \in P \setminus D, \\ ucB_R, & D = P, \quad R \in \mathcal{G} \text{ proper}, \quad c \in P \setminus R. \end{cases}$$

The second line is defined only when the required R exists; that proviso is exactly the exceptional case of Theorem 7.1. For every $m < \omega$ put

$$\Phi_m(\mathbf{u}) = g_D(u_0)B_D g_D(u_1)B_D \cdots B_D g_D(u_m).$$

Then

$$\Phi_m(\mathbf{u}) \preceq \Phi_m(\mathbf{v}) \iff (\forall i \leq m) u_i \preceq v_i.$$

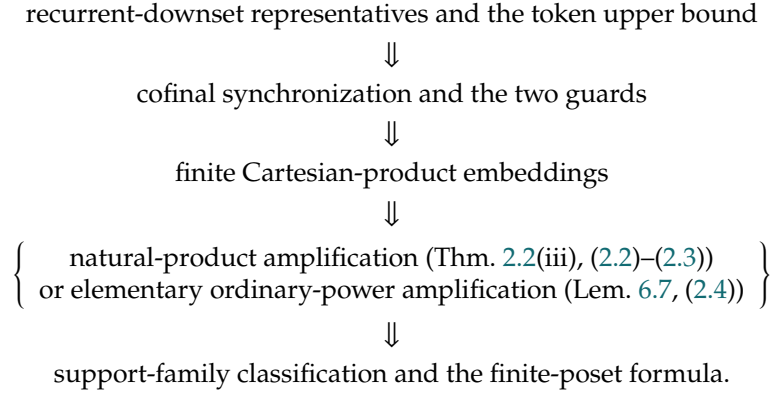
The proof is synchronization (Lemma 4.4 with Corollary 4.5), plus nonleakage (Lemma 6.2), plus cancellation (Lemmas 5.1 and 6.1). If $o(\mathcal{W}(P, \mathcal{G})) = H_q$ with $q = n + |\mathcal{G}|$, the equivalence yields

$$H_{q+1} = \sup_{m < \omega} H_q^{m+1} \leq o(\mathcal{W}(P, \mathcal{G} \cup \{D\})) \leq H_{q+1},$$

where the left equality may be read either through the natural powers of (2.2)–(2.3) or through the ordinary powers of (2.4) and Lemma 6.7, and the right inequality is Proposition 3.8. This is the complete lower-bound mechanism used in the article.

B A compact proof-dependency map

In outline, the argument has five layers:



The fourth layer is the only one at which the article offers two independent routes; either suffices for Theorem 1.1. The table below expands the same dependencies statement by statement.

Statement	Inputs	Role
Finite representation, Proposition 3.5	Finiteness of P ; recurrent labels; monotone concatenation	Identifies the full word quotient with a finite-token quotient.
Atom-count upper bound, Proposition 3.8	Finite-word formula; monotone surjection	Gives $o(\mathcal{W}(P, \mathcal{F})) \leq H_{n+ \mathcal{F} }$.
Cofinal localization, Lemma 4.1	Finitely many target tokens; strict increase; periodic recurrence	Finds the target block used cofinally by each source marker.
Synchronization, Lemma 4.4	Cofinal localization; no old block contains the new downset	Forces the i th new marker to correspond to the i th new marker.
Proper product embedding, Theorem 5.2	Synchronization; a letter outside the proper downset	Embeds every finite Cartesian power of the old stage.
Limit padding, Lemma 6.1	A proper old block; an excluded letter; final-letter cancellation	Produces an order-reflecting map into nonzero limit lengths.
Universal product embedding, Theorem 6.3	Synchronization; limit padding; the finite-bound contradiction	Adds the universal block without losing the product lower bound.
Family classification, Theorem 7.1	The two product maps; classical natural product; atom upper bound	Computes every selected family, including its sole exception.
Main theorem, Theorem 1.1	Full family has a proper downset when $n \geq 2$; finite representation	Gives $\omega^{\omega^{n+j(P)-2}}$ and the two boundary cases.
Unified product embedding, Theorem 6.5	The same synchronization, with one case-split guard map	Restates Theorems 5.2 and 6.3 as one equivalence.
Elementary amplification, Lemma 6.7 and Corollary 6.8	A maximal linear extension; the last-unequal-coordinate order	Replaces the natural-product import in the main line.
One-block normal form, Lemma 3.4	Last occurrence outside the recurrent downset	Supplies the admissible prefixes used by (10.3).
Equal-count rigidity, Lemma 10.2	Cofinal localization at a fixed block count	Whole-block containment; used only in Sections 10 and 11.3.
Exact-length strata, Propositions 10.1 and 10.3	Equal-count rigidity; natural sum and product	Supplementary; yields the warning (10.7).
Canonical representatives, Proposition 11.2	Normalized prefixes; equal-count rigidity; (10.3)	Makes the finite representation unique and effective.

C Minimal worked use of the code

The following example can be run with code/ on the Python import path. Downset masks 1, 2, and 3 represent $\{a\}$, $\{b\}$, and $\{a, b\}$.

```
from omega2 import (FinitePoset, letter, omega, embeds, normalize,
                    encode_product, theorem_value)

P = FinitePoset.antichain(2)
a, b = letter(0), letter(1)
A, B, U = omega(1), omega(2), omega(3)

assert embeds(P, (a, U), (U,)) # finite-prefix absorption
assert embeds(P, (A,), (U,)) # pure tail into universal tail
assert not embeds(P, (A, B), (U,)) # two cofinal blocks cannot share one
assert not embeds(P, (U,), (A, B)) # no eligible cofinal destination

assert normalize((a, U)) == (U,) # canonical quotient representative
assert normalize((b, A)) == (b, A) # b is outside the downset of A

# The unified guarded product map, introducing B over the family {A}.
print(encode_product(P, (1,), 2, (((), (a,))))
# (('f', 0), ('w', 2), ('f', 0), ('f', 0))

print(theorem_value(P))
# omega^(omega^4)

C = FinitePoset.chain(2)
print(theorem_value(C))
# omega^(omega^3)

print(theorem_value(P, family=(P.full,)))
# omega^(omega^1 + 1) = omega^(omega + 1)
```

The last output illustrates the exceptional universal-only family. Its maximal order type is much smaller than $H_3 = \omega^{\omega^2}$, even though the two finite letters and the universal block form three distinct formal tokens.

D Literature and reproducibility record

The chosen source was checked both in arXiv HTML and in the PDF rendering; Example 3.15 is on printed page 10, which is zero-based PDF page 9. Its version identifier is 2409.03199v2. The source of the open question, the two standard maximal-order-type references, and the distinction between proven classical inputs and the present proposed argument are recorded in notes/literature_audit.md.

The archive contains this source and its compiled PDF, a reusable references.bib, the symbolic module, the reproducible verification script, machine-readable test results, example and downset-count tables, a proof-audit note, and a README. It contains no third-party article PDFs, font files, checksum files, or LaTeX build intermediates. The file inventory is the following.

File	Purpose
article.tex	Complete editable source of this manuscript.
article.pdf	Compiled article.
references.bib	Bibliographic records, maintained alongside the embedded thebibliography.
build.sh	Three-pass pdflatex build script.
README.md	Scope, result list, reproduction, and library usage.

File	Purpose
code/omega2.py	Finite posets, downsets, exact token comparators, normalization, the guard maps, and the product encodings.
code/verify.py	Exhaustive and seeded checks, negative controls, and table generation.
data/verification.json, data/verification.txt data/test_summary.tex data/examples.csv data/ideal_counts.csv	Recorded output of the default verification run.
data/finite_posets.csv	Test-count macros used by this article. The computed finite-poset examples of Table 1. Downset-count histogram for naturally labelled posets of size at most five.
notes/proof_audit.md notes/literature_audit.md	The 50 naturally labelled posets through size four, individually, with predecessor and successor masks, downset counts, and the exponent predicted by (1.1). Hypotheses, critical steps, and known false shortcuts. Source, version, and literature-search scope.

References

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- [2] D. H. J. de Jongh and Rohit Parikh. Well-partial orderings and hierarchies. *Indagationes Mathematicae (Proceedings)*, 80(3):195–207, 1977; also cited as *Indagationes Mathematicae* 39. [https://doi.org/10.1016/1385-7258\(77\)90067-1](https://doi.org/10.1016/1385-7258(77)90067-1).
- [3] Diana Schmidt. Well-Partial Orderings and their Maximal Order Types. Habilitationsschrift, Heidelberg, 1979. Reprinted in P. Schuster, M. Seisenberger, and A. Weiermann, editors, *Well-Quasi Orders in Computation, Logic, Language and Reasoning*, Trends in Logic 53, pp. 351–391. Springer, 2020. https://doi.org/10.1007/978-3-030-30229-0_13.
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- [5] Alakh Dhruv Chopra and Fedor Pakhomov. *Well-quasi-orders on finite trees and transfinite sequences*. arXiv:2602.09830v1, February 10, 2026. <https://arxiv.org/abs/2602.09830v1>.